

An odd-prime obstruction for rational equal-weight quadrature

Research note, 30 September 2026

Theorem 1. *Let p be an odd prime. Suppose rational numbers $y_1, \dots, y_K$ form an equal-weight quadrature rule of degree at least $\max\{p-1, 3\}$ for the uniform probability measure on $[-1, 1]$. Then $K \geq 2p$. The nodes need not belong to $[-1, 1]$ and repetitions are permitted.*

Proof. Suppose $K < 2p$. The moment identities are

$$\sum_{i=1}^K y_i^s = \begin{cases} 0, & s \text{ odd}, \\ K/(s+1), & s \text{ even}, \end{cases} \quad 1 \leq s \leq \max\{p-1, 3\}.$$

Write v_p for the p -adic valuation.

First suppose all y_i are p -adically integral. The identity at $s = p-1$ forces $p \mid K$, so $K = p$. Reducing that identity modulo p , Fermat's theorem shows that the number q of unit nodes satisfies $q \equiv 1 \pmod{p}$. Since $0 \leq q \leq p$, exactly one node is a unit. The first moment then cannot vanish modulo p , a contradiction.

Otherwise let $a = -\min_i v_p(y_i) > 0$, and set $z_i = p^a y_i$. All z_i are p -adically integral and at least one is a unit. For $0 \leq s \leq p-2$ we have

$$\sum_{i: p \nmid z_i} z_i^s \equiv 0 \pmod{p}.$$

For $s = 1, \dots, p-2$ this follows from the scaled moment identities; their denominators $s+1$ are prime to p . For $s = 0$, first use $s = p-1$: its scaled right hand side $p^{a(p-1)}K/p$ is divisible by p , hence the number of units is divisible by p . It lies between 1 and $K < 2p$, so it is precisely p .

For $b \in \mathbb{F}_p^*$ let N_b count the unit nodes with residue b . The displayed congruences give

$$\sum_{b \in \mathbb{F}_p^*} N_b b^s = 0 \quad (0 \leq s \leq p-2).$$

The Vandermonde matrix on the $p-1$ distinct nonzero residues is invertible. Thus every N_b is divisible by p . Since their sum is p , all p unit nodes have a common nonzero residue b .

If $p = 3$, for any p -adically integral $z \equiv b \pmod{p}$, the binomial theorem gives $z^p \equiv b^p \pmod{p^2}$. Nonunit nodes have $z_i^p \equiv 0 \pmod{p^2}$. Consequently

$$\sum_i z_i^p \equiv pb^p \not\equiv 0 \pmod{p^2},$$

contradicting the vanishing third moment.

If $p \geq 5$, write the unit nodes as $z_i = b + pt_i$, and let $T = \sum_{i: p \nmid z_i} t_i \pmod{p}$. The scaled second and third moments are both zero modulo p^2 : the second is $Kp^{2a}/3$, and the third is zero. Nonunit nodes contribute zero modulo p^2 to both moments. Expanding the unit contributions gives

$$0 \equiv pb^2 + 2pbT \pmod{p^2}, \quad 0 \equiv pb^3 + 3pb^2T \pmod{p^2}.$$

Since b is nonzero modulo p , these imply $b + 2T = b + 3T = 0$ in $\mathbb{F}_p$, a contradiction. $\square$

In particular, rational degree-three rules need at least six nodes, and rational degree-four rules need at least ten nodes. The first bound is sharp: the multiset $(-1, 0, 0, 0, 0, 1)$ is the six-node Simpson rule. These bounds complement the stronger asymptotic bound $K \geq 2^{\lfloor r/2 \rfloor}$ proved in `../rational_designs/two_adic_lower_bound.tex`. Affine rescaling transfers the statement to the uniform measure on $[0, 1]$.